RAG CAPSTONE

Wikipedia research assistant

Answers grounded in a saved article collection

Design, evaluation, and engineering trade-offs



THE PROBLEM

Example research questions

ONE FACT

“What overall pick was Jeremiyah Love in the 2026 NFL draft?”

Retrieve the selection and its supporting passage.

A CONNECTION ACROSS ARTICLES

“What is the capital of the state that the actor who played Cooter Davenport in The Dukes of Hazzard represented in Congress?”

Support the actor, represented state, and capital.

A COMPARISON ACROSS DRAFTS

“what spot did Jeremiyah Love get drafted and compare it to the same round of 5 older drafts for Arizona.”

Resolve the round and draft years, then support every comparison.

A DATE AND AN AMOUNT

“When and for how much did the Brady Bunch house sell?”

Support both the sale date and the price.

THE SOLUTION

System workflow

01. Load and index

Extract local HTML text, embed articles, preserve source IDs.

02. Choose a path

Check a saved answer, use direct retrieval, or run the agent.

03. Retrieve

Combine BM25 keyword matches with vector similarity.

04. Answer with sources

Generate from bounded excerpts and cite article IDs.

RUN SETTINGS

Retrieval choices and context limits

Choice	Trade-off
Equal-weight keyword and vector scores	Combines matching signals; weights need evaluation.
Local article-link traversal	Adds related evidence within the saved collection.
3 action rounds, 1.5M context characters	Bounds work but can stop before finding enough evidence.
Whole articles, no per-article cap	More evidence per article, with greater token usage.

EVALUATION

Four questions one fresh run

4

questions rerun

- Draft pick and Brady house: requested facts supported.
- Arizona: one older draft found; four still missing.
- Dukes: actor and state supported; capital claim unsupported.

Draft pick

What overall pick was Jeremiyah Love in the 2026 NFL draft?

Latency	Model calls	Chat tokens	Est. chat cost
15.57 s	**5**	**150,797**	**$0.07828**

AI RESPONSE EXCERPT

> “Jeremiyah Love was the 3rd overall pick in the 2026 NFL draft. [2026_NFL_draft]”

EVIDENCE REVIEW

3rd overall is supported by the retained draft table.

Stopped: Planner chose to answer after finding the draft pick.

Arizona draft comparison

what spot did Jeremiyah Love get drafted and compare it to the same round of 5 older drafts for Arizona.

Latency	Model calls	Chat tokens	Est. chat cost
17.76 s	**5**	**188,885**	**$0.07070**

AI RESPONSE EXCERPT

> “Jeremiyah Love was drafted **3rd overall** by the Arizona Cardinals in the **2026 NFL draft** [2026_NFL_draft].”

EVIDENCE REVIEW

Two picks supported; four older drafts still missing, as the answer acknowledges.

Stopped: Planner chose to answer despite the incomplete comparison.

The Dukes of Hazzard

What is the capital of the state that the actor who played Cooter Davenport in The Dukes of Hazzard represented in Congress?

Latency	Model calls	Chat tokens	Est. chat cost
20.31 s	**6**	**181,083**	**$0.10552**

AI RESPONSE EXCERPT

> “The capital of Georgia is Atlanta [Ben_Jones_(American_actor_and_politician)].”

EVIDENCE REVIEW

Actor and state supported; the cited Ben Jones article does not establish the capital.

Stopped: Planner returned an unknown or unavailable action.

The Brady Bunch house

When and for how much did the Brady Bunch house sell?

Latency	Model calls	Chat tokens	Est. chat cost
26.85 s	**7**	**494,006**	**$0.23979**

AI RESPONSE EXCERPT

> “The Brady Bunch house was bought on September 11, 2023, for $3.2 million. [The_Brady_Bunch]”

EVIDENCE REVIEW

September 11, 2023 and $3.2 million are both supported by the retained article.

Stopped: Planner chose to answer after finding the sale details.

SAFETY AND COST

Each extra check has a cost

- Input cleaning and XML escaping protect prompt boundaries.
- Cached answers still require validation.
- Filtering, routing, and quality checks add model calls.
- Chat cost estimates exclude embedding costs.



LIMITATIONS AND FUTURE WORK

Next improvements

01. **Retrieve passages**

    Find the relevant text inside long articles.

02. **Check each claim**

    Require supporting evidence for every requested fact.

03. **Strengthen evaluation**

    Complete the grading rubric and expand the question set.

04. **Handle invalid decisions**

    Review planner actions and failed JSON checks.

ENGINEERING CONCLUSION

Retrieval, evidence, and verification must work together

Search makes evidence available.

Review establishes what that evidence supports.

github.com/sterlingDB/mit-rag-class